WORKED SOLUTION

# STEP I 2014, Question 3

Pure • Curve sketching

**STEP I 2014, Question 3**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q3

## The key idea

Both integrals are elementary, so the condition is just a polynomial equation in $a$ and $b$. The work is in spotting the common factor $b-a$ (which is non-zero, so we may cancel it) and, in part (iii), in noticing that the leftover expression is symmetric enough to be written neatly in terms of the sum $b+a$ and the difference $b-a$.

## Solution

Evaluating the integrals, the condition is

$$\frac{b^3-a^3}{3} = \left(\frac{b^2-a^2}{2}\right)^2 = \frac{(b-a)^2(b+a)^2}{4}. \qquad (*)$$

Since $b^3-a^3 = (b-a)(b^2+ab+a^2)$ and $b-a>0$, we can divide $(*)$ by $b-a$ to get

$$4(b^2+ab+a^2) = 3(b-a)(b+a)^2. \qquad (**)$$

### Part (i)

With $a=0$, $(**)$ becomes $4b^2 = 3b^3$. As $b>0$ we can divide by $b^2$, giving

$$b = \tfrac{4}{3}.$$

### Part (ii)

With $a=1$, $(**)$ becomes

$$4(b^2+b+1) = 3(b-1)(b+1)^2 = 3(b^3+b^2-b-1),$$

that is, $3b^3+3b^2-3b-3-4b^2-4b-4=0$, or

$$3b^3-b^2-7b-7=0,$$

as required.

Now let $f(x) = 3x^3 - x^2 - 7x - 7$, so that

$$f'(x) = 9x^2 - 2x - 7 = (9x + 7)(x - 1).$$

The stationary points are at $x = -\frac{7}{9}$ (a local maximum, since the cubic has positive leading coefficient) and $x = 1$ (a local minimum), with

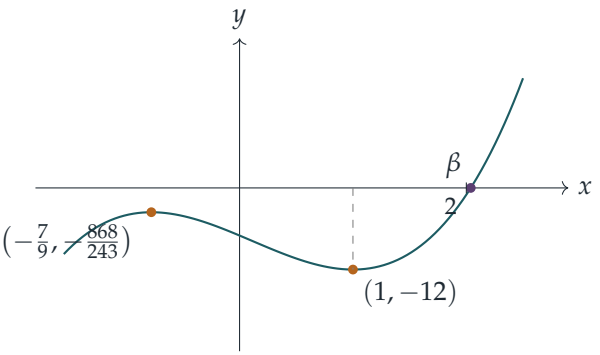
$$f(1) = 3 - 1 - 7 - 7 = -12, \qquad f\left(-\tfrac{7}{9}\right) = \frac{-343 - 147 + 1323 - 1701}{243} = -\frac{868}{243}.$$

**Ask yourself** How does knowing the value at the local maximum tell us how many roots there are?

The local maximum value is negative. For $x \leq 1$, $f$ increases up to $x = -\frac{7}{9}$ and then decreases, so $f(x) \leq f(-\frac{7}{9}) < 0$ there: no roots with $x \leq 1$. For $x > 1$, $f$ is strictly increasing (both factors of $f'$ are positive) and $f(x) \to \infty$, so it crosses zero exactly once. Hence there is exactly one real root, and

$$f(2) = 24 - 4 - 14 - 7 = -1 < 0, \qquad f(3) = 81 - 9 - 21 - 7 = 44 > 0,$$

so by continuity it lies between 2 and 3.




The sketch shows the curve $y = 3x^3 - x^2 - 7x - 7$: both turning points lie below the $x$-axis, so it crosses only once, at $\beta \approx 2.04$.

**Part (iii)**

With $p = b + a$ and $q = b - a$ we have $p^2 = b^2 + 2ab + a^2$ and $q^2 = b^2 - 2ab + a^2$, so

$$3p^2 + q^2 = 4b^2 + 4ab + 4a^2 = 4(b^2 + ab + a^2).$$

The left-hand side of $(**)$ is therefore $3p^2 + q^2$, and the right-hand side is $3qp^2$. So

$$3p^2 + q^2 = 3p^2 q,$$

as required. Rearranging, $q^2 = 3p^2(q-1)$. Since $q > 0$, the left side is positive, so $q \neq 1$ and

$$p^2 = \frac{q^2}{3(q-1)}.$$

For the deduction:

- *Lower bound.* $q^2 > 0$ and $p^2 > 0$ (as $p = b + a > 0$), so $3(q-1) = q^2/p^2 > 0$, i.e. $q > 1$.
- *Upper bound.* Since $a \geq 0$, we have $p = q + 2a \geq q > 0$, so $p^2 \geq q^2$. Hence

  $$\frac{q^2}{3(q-1)} \geq q^2 \iff 1 \geq 3(q-1) \iff q \leq \tfrac{4}{3},$$

  where we divided by $q^2 > 0$ and multiplied by $3(q-1) > 0$, both of which preserve the inequality.

Therefore $1 < b - a \leq \frac{4}{3}$, with equality exactly when $a = 0$, which is part (i).

## Checks

- Part (i) fits part (iii): $a = 0$ gives $q = b = \frac{4}{3}$, the top of the range.
- Part (ii) fits too: the root is $b \approx 2.0390$, so $q = b - 1 \approx 1.039$, inside $(1, \frac{4}{3}]$. Evaluating both integrals over $[1, 2.0390]$ gives equal values (about 2.4924) to many decimal places.
- Conversely, any $q$ in $(1, \frac{4}{3}]$ gives $p = q/\sqrt{3(q-1)} \geq q$ and so a valid $a = \frac{1}{2}(p-q) \geq 0$; the bound is sharp.

## Common slips

- Forgetting to cancel $b - a$ in part (ii), leaving a quartic with the spurious root $b = 1$.
- Showing only that $f(2) < 0 < f(3)$: this proves there is *a* root in $(2, 3)$, but not that it is the only real root. You need the turning points too.
- In part (iii), dividing an inequality by $q - 1$ before establishing that $q - 1 > 0$.

## Beyond STEP

The Cauchy–Schwarz inequality for integrals gives $\left(\int_a^b x\,\mathrm{d}x\right)^2 \leq (b-a)\int_a^b x^2\,\mathrm{d}x$, with equality only when the integrand $x$ is constant, which it is not. Combined with the given condition this says $\int_a^b x^2\,\mathrm{d}x < (b-a)\int_a^b x^2\,\mathrm{d}x$, so $b - a > 1$: the lower bound of part (iii) in one line. Cauchy–Schwarz, in its many forms, is one of the most used tools in first-year analysis; Bartle and Sherbert's *Introduction to Real Analysis* develops the Riemann integral and inequalities like this one carefully.

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